

Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

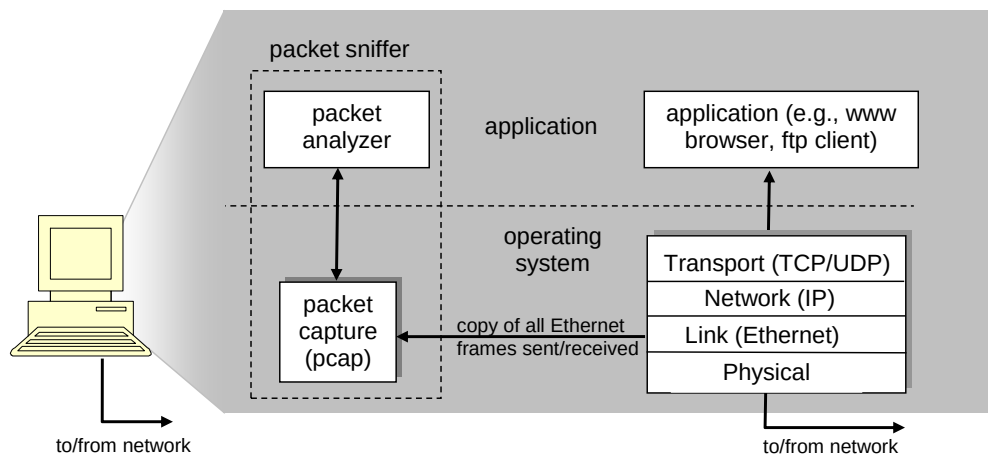


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

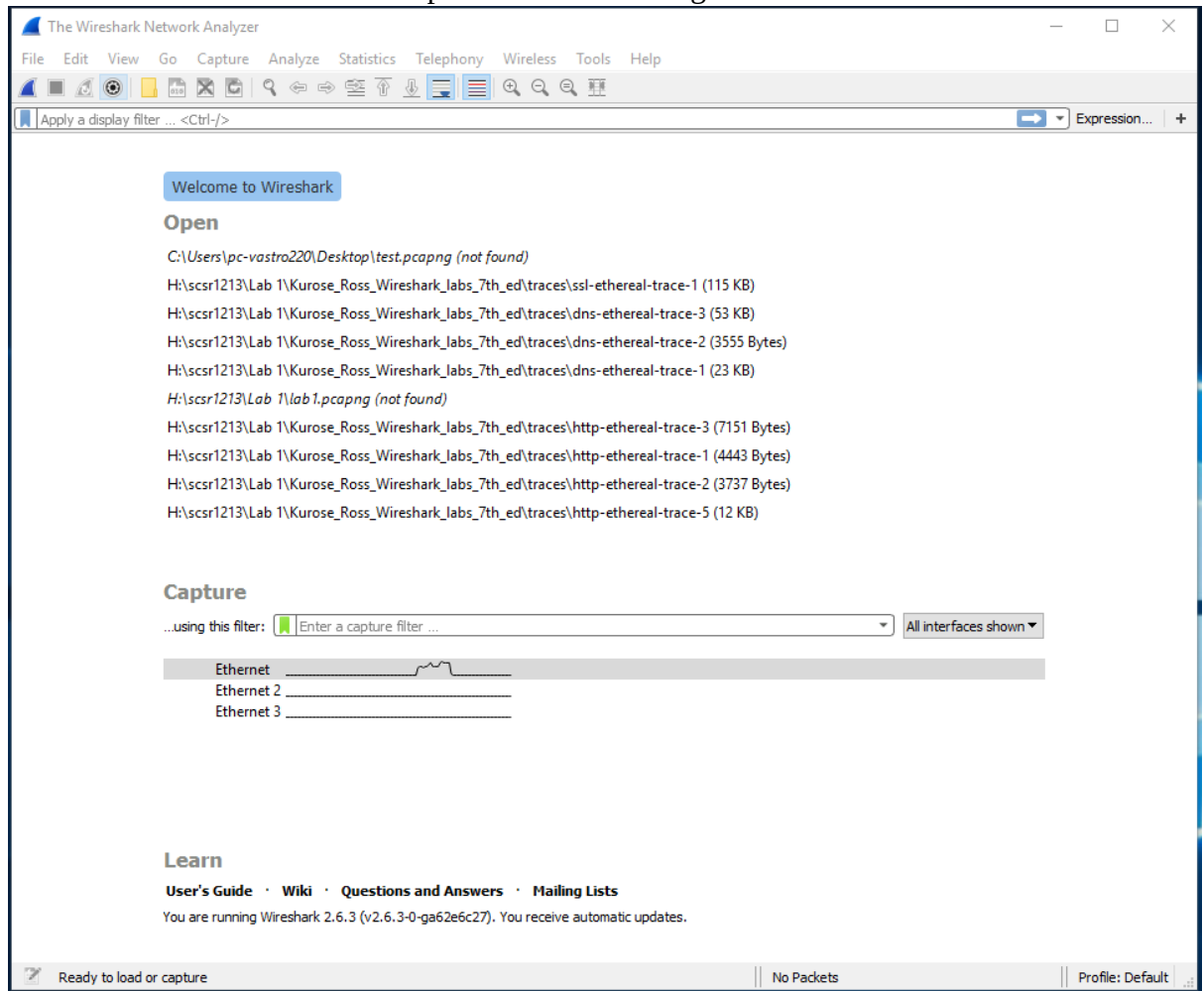


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

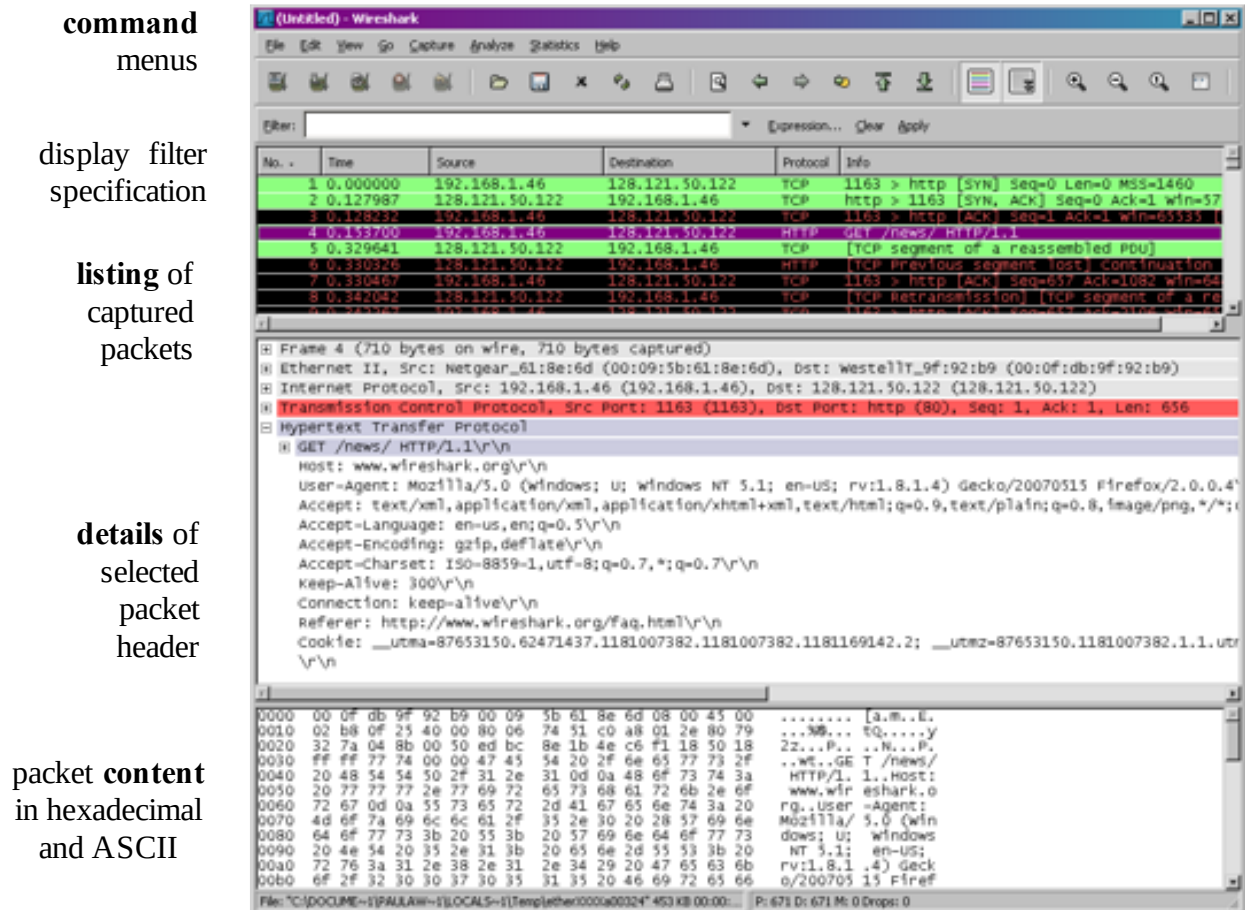


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

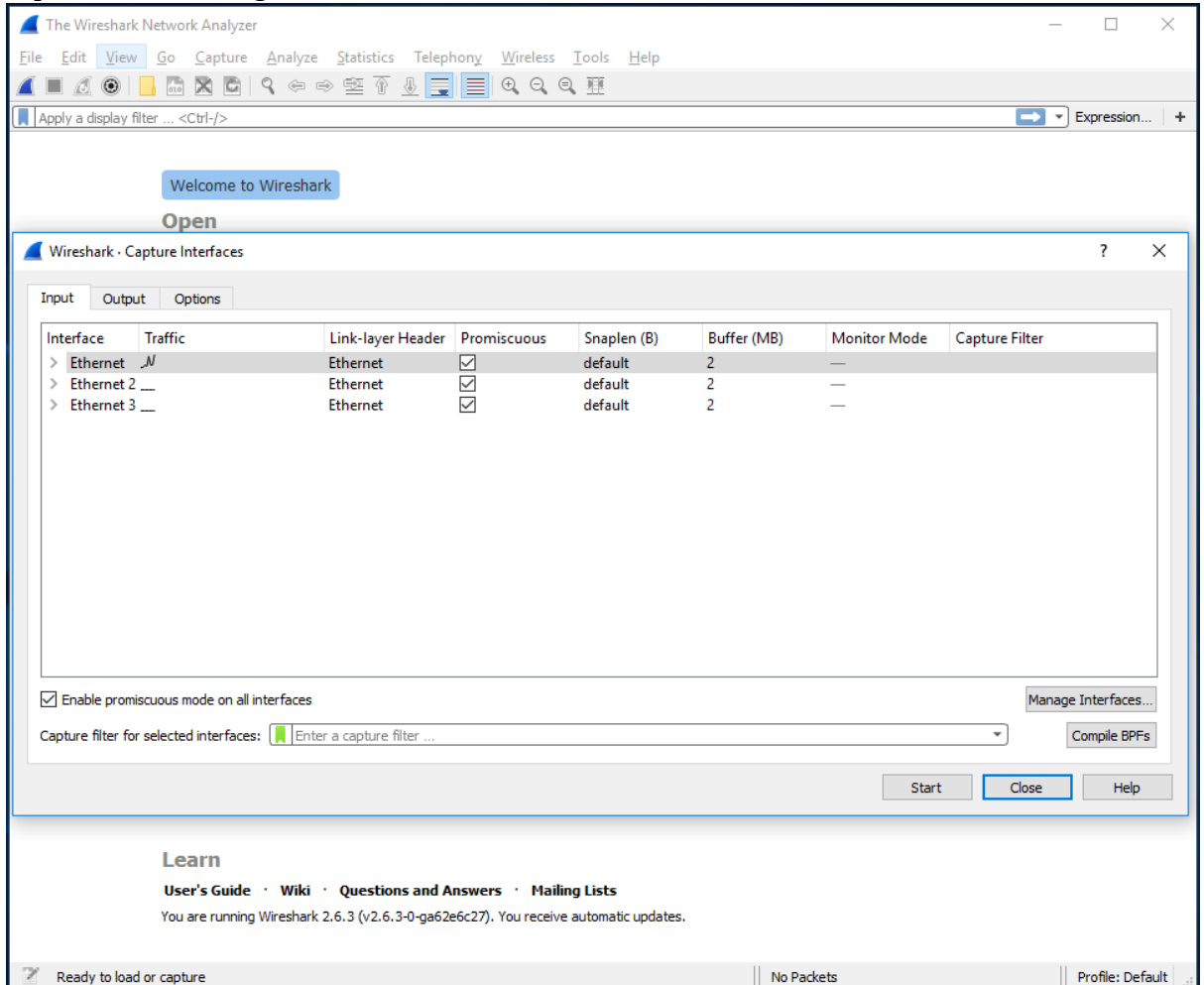


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

The image shows the Wireshark interface with a packet capture in progress. The top pane displays a list of 90 captured packets. The bottom pane shows the details of the selected packet (No. 90), which is an ARP request from 10.60.80.78 to 255.255.255.255.

No.	Time	Source	Destination	Protocol	Length	Info
80	4.169920	Dell_87:3b:e4	Broadcast	ARP	60	Who has 10.60.83.65? Tell 10.60.80.70
81	4.344789	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
82	4.346792	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
83	4.347075	10.60.82.204	10.60.83.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
84	4.351986	HewlettP_15:15:2b	Broadcast	ARP	60	Who has 10.60.80.15? Tell 10.60.80.167
85	4.479278	Cisco_0d:7b:04	Spanning-tree-(for-...	STP	119	MST. Root = 0/10/00:23:04:ee:be:03 Cost = 20004 Port
86	4.530418	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
87	4.530808	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
88	4.700996	Dell_75:07:b3	Broadcast	ARP	60	Who has 10.60.80.162? Tell 10.60.80.64
89	4.767952	10.60.82.204	10.60.83.255	NBNS	92	Name query NB MSI-PC<1c>
90	4.995012	10.60.80.78	255.255.255.255	DB-LSP...	176	Dropbox LAN sync Discovery Protocol

Frame 1: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface 0
 Ethernet II, Src: Dell_23:e3:3a (00:24:e8:23:e3:3a), Dst: Cisco_d5:79:ff (00:14:6a:d5:79:ff)
 Internet Protocol Version 4, Src: 10.60.82.216, Dst: 52.230.84.0
 Transmission Control Protocol, Src Port: 49724, Dst Port: 443, Seq: 1, Ack: 1, Len: 53
 Secure Sockets Layer

0000 00 14 6a d5 79 ff 00 24 e8 23 e3 3a 08 00 45 00 ..j.y.\$ #.:..E.
 0010 00 5d 64 e9 40 00 80 06 00 00 0a 3c 52 d8 34 e6 ..]d.@...<R.4.
 0020 54 00 c2 3c 01 bb 11 3f 6d 43 3e d4 b9 24 50 18 T...<...? mC>..\$P.
 0030 01 01 e6 49 00 00 17 03 01 00 30 07 75 ee af 3c ...I.....0-u...<
 0040 42 81 47 9e 52 29 c3 be ea 2e 88 ac 28 ed bf e6 B.G.R)...((...
 0050 20 36 00 a7 ad f6 ff 8a b0 6d 63 a3 c7 a4 68 ca 6.....mc...h.
 0060 fd 6d b3 05 36 73 0e 3c 54 bc 07 ..m...6s.<T..

Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

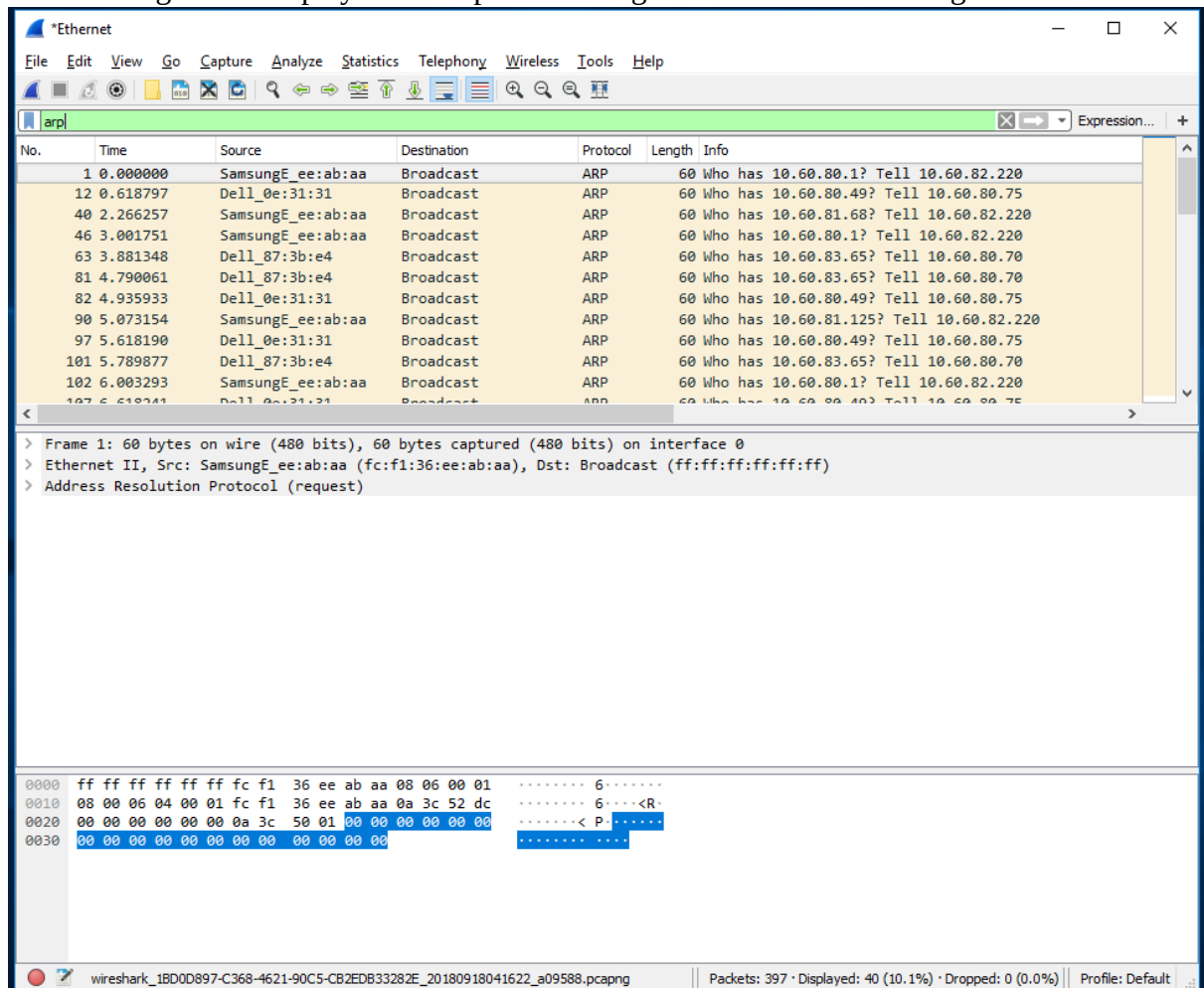


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

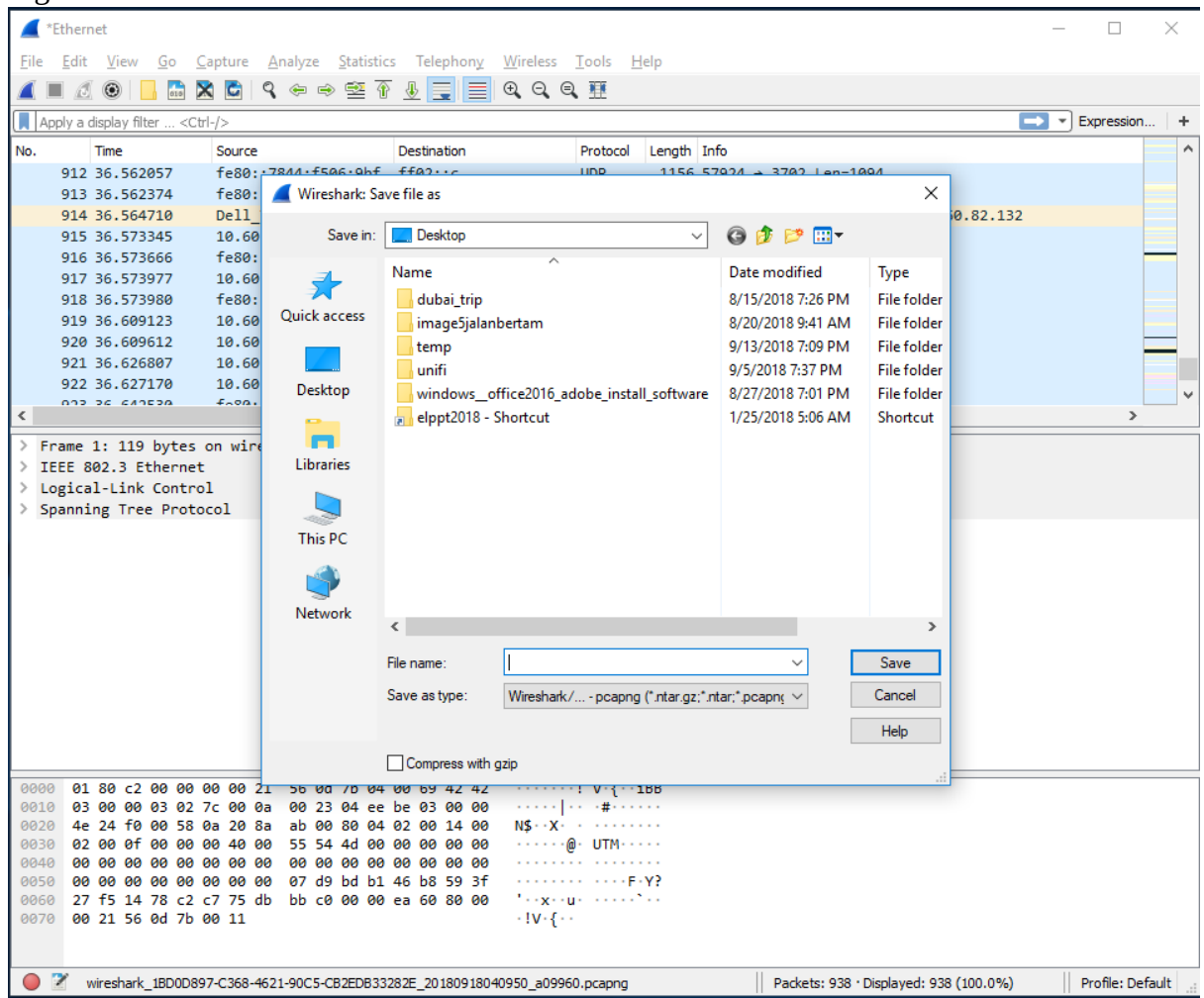


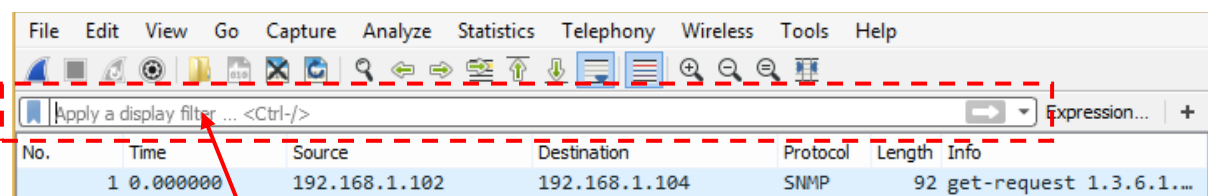
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

2. What is the IP address of the client computer?

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

3. What is the IP address of the gaia.cs.umass.edu server?

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

4. How many bytes of content are being returned to client browser?

```

v Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
    Server: Apache/2.0.40 (Red Hat Linux)\r\n
    Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n
    ETag: "1bfed-49-79d5bf00"\r\n
    Accept-Ranges: bytes\r\n
    Content-Length: 73\r\n
    Keep-Alive: timeout=10, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=ISO-8859-1\r\n

```

73 bytes

```

v Hypertext Transfer Protocol
  > HTTP/1.1 404 Not Found\r\n
    Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
    Server: Apache/2.0.40 (Red Hat Linux)\r\n
    Vary: accept-language\r\n
    Accept-Ranges: bytes\r\n
    Content-Length: 1071\r\n
    [Content length: 1071]
    Keep-Alive: timeout=10, max=99\r\n
    Connection: Keep-Alive\r\n

```

1071 bytes

5. What is the status code returned from the server to client browser?

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

```
▼ Hypertext Transfer Protocol
  > GET /ethereal-labs/lab2-2.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
    Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image
    Accept-Language: en-us, en;q=0.50\r\n
    Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
    Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
    Keep-Alive: 300\r\n
    Connection: keep-alive\r\n
    \r\n
    [Response in frame: 10]
    [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-2.html]
```

No, there is no “IF-MODIFIED-SINCE” line in the HTTP GET.

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

Yes, the server explicitly returns the contents of the file. This is shown by the status code 200 and the response phrase “OK” which prove that it returns the contents of the file successfully.

```
▼ Line-based text data: text/html (10 lines)
  \n
  <html>\n
  \n
  Congratulations again! Now you've downloaded the file lab2-2.html. <br>\n
  This file's last modification date will not change. <p>\n
  Thus if you download this multiple times on your browser, a complete copy <br>\n
  will only be sent once by the server due to the inclusion of the IF-MODIFIED-SINCE<br>\n
  field in your browser's HTTP GET request to the server.\n
  \n
  </html>\n
```

It shows the contents of the file that will return.

- Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes, the captured HTTP messages of the second HTTP GET request from the browser to the server contain the “IF-MODIFIED-SINCE:” line. The information follows the “IF-MODIFIED-SINCE:” header is “Tue, 23 Sep 2003 05:35:00 GMT\r\n”.

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

Hypertext Transfer Protocol

```

> GET /ethereal-labs/lab2-2.html HTTP/1.1\r\n
Host: gaia.cs.umass.edu\r\n
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image.
Accept-Language: en-us, en;q=0.50\r\n
Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
Keep-Alive: 300\r\n
Connection: keep-alive\r\n
If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n
If-None-Match: "1bfef-173-8f4ae900"\r\n
Cache-Control: max-age=0\r\n
\r\n

```

- What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

The HTTP status code from the server in response to this second HTTP GET is 304 and the phrase returned is “Not Modified”. The server did not return the contents of the file. This is because the server detects the file does not change since last access and send the status code 304 to inform the browser to use the previous content in the cache.

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

No.	Time	Source	Destination	Protocol	Length	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.260813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

There are 3 HTTP GET request messages send by the client browser.

2. To which Internet addresses were these GET requests sent?

No.	Time	Source	Destination	Protocol	Length	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.260813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

3. How many bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

```
▼ Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Server: Netscape-Enterprise/3.6 SP3\r\n
    Date: Sun, 21 Sep 2003 06:00:35 GMT\r\n
    Content-type: image/gif\r\n
    Etag: "6fc149-d1d-3ef0b3f8"\r\n
    Last-modified: Wed, 18 Jun 2003 18:48:24 GMT\r\n
  > Content-length: 3357\r\n
    Accept-ranges: bytes\r\n
    Connection: keep-alive\r\n
    \r\n
```

3357 bytes

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

```
▼ Hypertext Transfer Protocol
  > HTTP/1.0 200 Document follows\r\n
    Date: Tue, 23 Sep 2003 05:38:44 GMT\r\n
    Server: NCSA/1.5.2\r\n
    Last-modified: Tue, 23 Sep 2003 04:56:38 GMT\r\n
    Content-type: image/jpeg\r\n
  > Content-length: 15642\r\n
    [Content length: 15642]
    \r\n
```

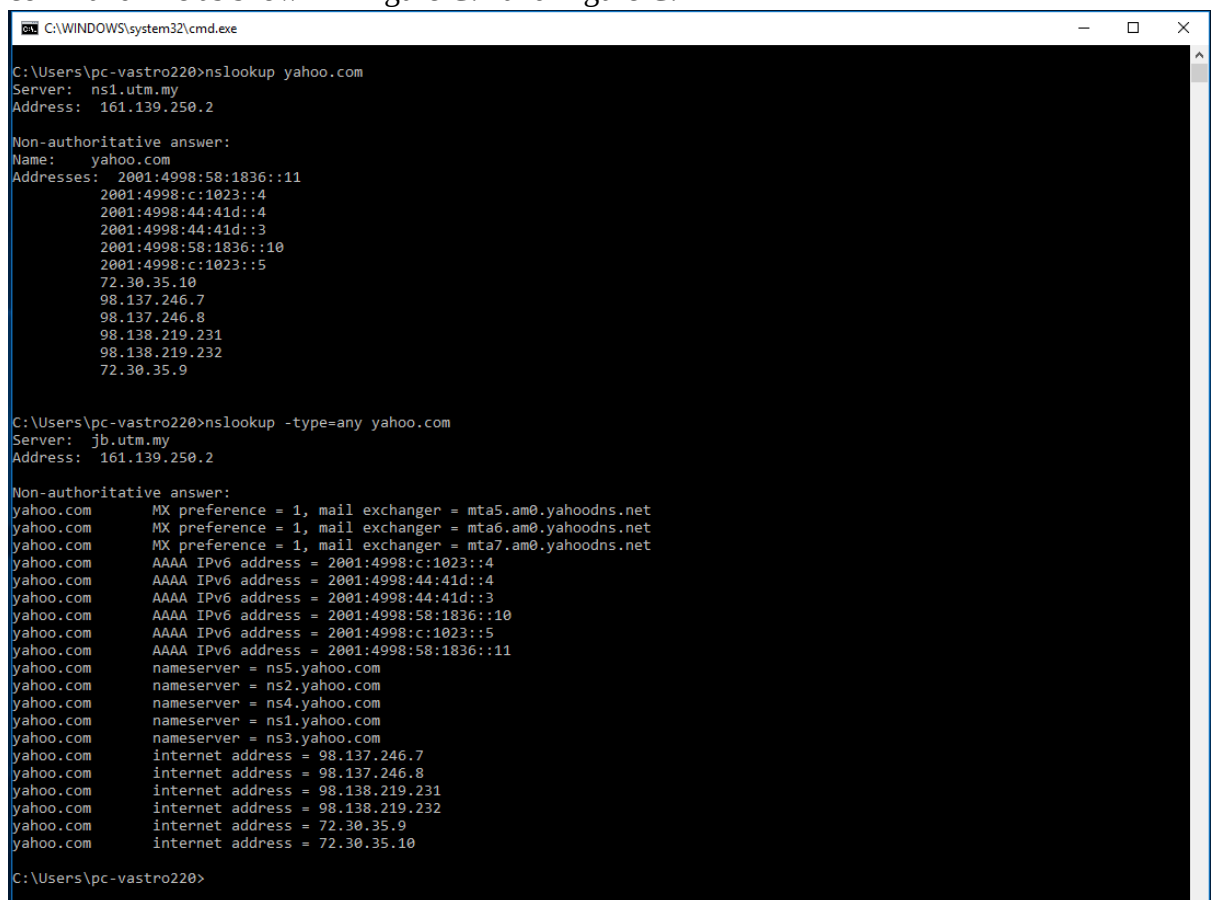
15642 bytes

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

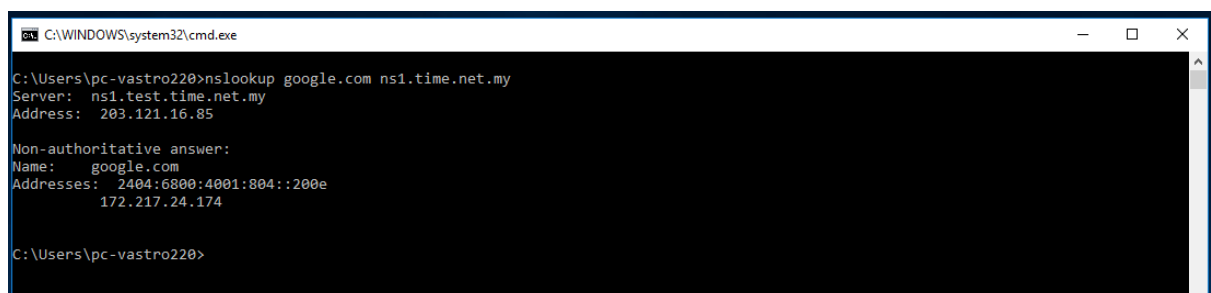
Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: jlb.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

```
C:\Users\Evelyn Goh>nslookup www.microsoft.com
Server: ns3.utm.my
Address: 161.139.168.168

Non-authoritative answer:
Name: e13678.dscb.akamaiedge.net
Addresses: 2001:f40:a:c8e::356e
           2001:f40:a:c84::356e
           2001:f40:a:c82::356e
           2001:f40:a:c81::356e
           2001:f40:a:c87::356e
           211.25.121.218
Aliases: www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

```
C:\Users\Evelyn Goh>nslookup www.microsoft.com
Server: ns3.utm.my
Address: 161.139.168.168

Non-authoritative answer:
Name: e13678.dscb.akamaiedge.net
Addresses: 2001:f40:a:c8e::356e
           2001:f40:a:c84::356e
           2001:f40:a:c82::356e
           2001:f40:a:c81::356e
           2001:f40:a:c87::356e
           211.25.121.218
Aliases: www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all

```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
Command: ipconfig /displaydns

```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

-----
aimsweb.utm.my
Record Name . . . . . : aimsweb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

-----
my.utm.my
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.
Command: ipconfig /flushdns

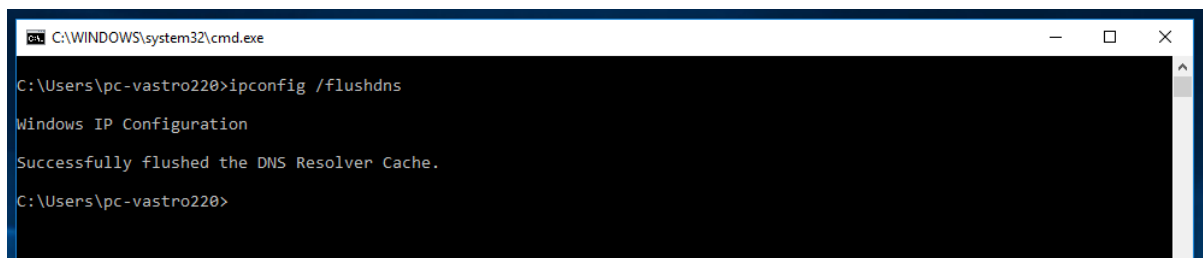


Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.
- Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco_fc:f0:de	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/0/00:01:96:45:05:9a Cost = 12 Port = 0x802d
2	0.148791	00000004.0001e62225...	00000004.ffffffffff...	IPX SAP	113	General Response
3	0.374081	Cisco_83:e4:54	Broadcast	ARP	60	Who has 128.238.38.248? Tell 128.238.38.2
4	1.981736	00000004.0001e62cea...	00000004.ffffffffff...	IPX SAP	113	General Response
5	1.999786	Cisco_fc:f0:de	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/0/00:01:96:45:05:9a Cost = 12 Port = 0x802d
6	2.031956	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
7	2.527474	Cisco_83:e4:54	Broadcast	ARP	60	Who has 128.238.38.38? Tell 128.238.38.2
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1

Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23

```

0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 58
Identification: 0x229e (8862)
> 000. .... = Flags: 0x0
... 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0xd281 [validation disabled]
[Header checksum status: Unverified]
Source Address: 128.238.38.160
Destination Address: 128.238.29.23
[Stream index: 1]

```

UDP (Query)

Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160

```

0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 90
Identification: 0xd595 (54677)
> 000. .... = Flags: 0x0
... 0000 0000 0000 = Fragment Offset: 0
Time to Live: 126
Protocol: UDP (17)
Header Checksum: 0x216a [validation disabled]
[Header checksum status: Unverified]
Source Address: 128.238.29.23
Destination Address: 128.238.38.160
[Stream index: 1]

```

UDP (Response)

2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

▼ User Datagram Protocol, Src Port: 3163, Dst Port: 53
Source Port: 3163
Destination Port: 53
Length: 38
Checksum: 0x8acb [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 1]

DNS (Query message)

▼ User Datagram Protocol, Src Port: 53, Dst Port: 3163
Source Port: 53
Destination Port: 3163
Length: 70
Checksum: 0xb0ba [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 2]

DNS response message

3. To what IP address is the DNS query message sent? Add screenshots in your answer.

No.	Time	Source	Destination	Protocol	Length	Info
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

```

v Domain Name System (query)
  Transaction ID: 0x006e
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  v Queries
    > www.ietf.org: type A, class IN
    [Response In: 9]

```

The “Type” of DNS query is Type A. The query message does not contain any “answers”.

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

The DNS response message are provided 2 “answers” which is shown in Figure C.3.5.1. Each of these answers contain the domain name, type of the address, the class, IP address, time to live and data length.

```

v Domain Name System (response)
  Transaction ID: 0x006e
  > Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 2
  Authority RRs: 0
  Additional RRs: 0
  v Queries
    > www.ietf.org: type A, class IN
  v Answers
    > www.ietf.org: type A, class IN, addr 132.151.6.75
    > www.ietf.org: type A, class IN, addr 65.246.255.51
    [Request In: 8]
    [Time: 0.000844000 seconds]
  v Answers
    v www.ietf.org: type A, class IN, addr 132.151.6.75
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 132.151.6.75
    v www.ietf.org: type A, class IN, addr 65.246.255.51
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 65.246.255.51

```

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

No.	Time	Source	Destination	Protocol	Length	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

No.	Time	Source	Destination	Protocol	Length	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM

No, the host does not issue new DNS queries before retrieving each image. There is no new DNS queries issue by host between line number 10 and number 27 before retrieving image in line number 28.

- Open packet trace file dns-trace-2 for nslookup.
 - We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
 - Answer the following questions.
8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

The destination port for the DNS query message is 53 and the source port of DNS response message is 53.

No.	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY

- v User Datagram Protocol, Src Port: 3742, Dst Port: 53
 - Source Port: 3742
 - Destination Port: 53
 - Length: 37
 - Checksum: 0x5890 [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 3]
 - [Stream Packet Number: 1]
- v User Datagram Protocol, Src Port: 53, Dst Port: 3742
 - Source Port: 53
 - Destination Port: 3742
 - Length: 162
 - Checksum: 0xa318 [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 3]
 - [Stream Packet Number: 2]

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

The IP address of the DNS query message sent is 128.238.29.22. This IP address is different from the default local DNS server.

No.	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY

```

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : MediaTek Wi-Fi 6E MT7922 (RZ616) 160MHz Wireless LAN Card
Physical Address. . . . . : 30-03-C8-38-3B-AF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::15f1:b09a:e196:9cff%9(Preferred)
IPv4 Address. . . . . : 10.205.106.247(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Lease Obtained. . . . . : Thursday, 19 December, 2024 3:07:06 PM
Lease Expires . . . . . : Thursday, 19 December, 2024 11:25:26 PM
Default Gateway . . . . . : 10.205.101.250
DHCP Server . . . . . : 161.139.100.238
DHCPv6 IAID . . . . . : 103809992
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-88-63-11-30-03-C8-38-3B-AF
DNS Servers . . . . . : 161.139.168.168
NetBIOS over Tcpip. . . . . : Enabled
  
```

10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

The “Type” of DNS query is Type A. The query does not contain any “answers”.

```
✓ Domain Name System (query)
  Transaction ID: 0x0003
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  ✓ Queries
    > www.mit.edu: type A, class IN
    [Response In: 20]
```

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

The DNS response message provided 1 “answer” as shown in Figure C.3.11.1. Each of these answers contain the domain name, the type of address, the class, time to live, data length and IP address.

```
✓ Domain Name System (response)
  Transaction ID: 0x0003
  > Flags: 0x8580 Standard query response, No error
  Questions: 1
  Answer RRs: 1
  Authority RRs: 3
  Additional RRs: 3
  ✓ Queries
    > www.mit.edu: type A, class IN
  ✓ Answers
    > www.mit.edu: type A, class IN, addr 18.7.22.83
  > Authoritative nameservers
  > Additional records
  [Request In: 19]
  [Time: 0.016757000 seconds]

✓ Answers
  ✓ www.mit.edu: type A, class IN, addr 18.7.22.83
    Name: www.mit.edu
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 60 (1 minute)
    Data length: 4
    Address: 18.7.22.83
  > Authoritative nameservers
```